

Quantum Mechanics from Stochastic Defect Diffusion on the 24-Cell Lattice

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We derive the Schrödinger equation from the stochastic diffusion of cone defects on the 24-cell lattice. We show that the wave function emerges as the probability amplitude for defect configurations, and the complex structure of quantum mechanics is shown to be a consequence of the spatial time-definition cycle.

I. INTRODUCTION

Quantum mechanics is often postulated as a set of axioms. We derive these axioms from the stochastic motion of lattice defects on the 24-cell manifold.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi \quad (\text{emergent Schrödinger equation}) \quad (1)$$

$$P(x, t) = |\psi(x, t)|^2 \quad (\text{Born rule from defect counting}) \quad (2)$$

III. EMERGENCE OF THE WAVE FUNCTION

By mapping the diffusion of defects under the spatial time cycle, we show that the diffusion equation maps to the Schrödinger equation, deriving the Born rule.

IV. CONCLUSION

Quantum mechanics is shown to be an emergent statistical description of the discrete spacetime lattice, bridging quantum and classical physics.

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- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

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